

Quantum Computing for Computer Scientists

Math Homework

Due date: January 23, 2024

Problem 1

Given the matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

- A) Show that they are hermitian and unitary.
- B) Compute the eigenvalues and eigenvectors of these matrices.

Problem 2

- A) Express the states $|+\rangle = H|0\rangle$ and $|-\rangle = H|1\rangle$ in the computational basis. (where H is the matrix from the previous problem)
- B) Show that $(|+\rangle, |-\rangle)$ is an orthonormal basis for a 2-dimensional Hilbert space.
- C) Express the 2-qubit state $|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ in the $(|++\rangle, |+-\rangle, |-+\rangle, |--\rangle)$ basis².

Problem 3

The trace of an operator A , acting on some vector space V , can be written as $Tr(A) = \sum_{i \in B} \langle i | A | i \rangle$, where B is an orthonormal basis of V . When we write $Tr(A) = \sum_i A_{ii}$ we are considering the canonical basis. Show that the trace function is independent of the choice of the basis. Concretely, show that for an *arbitrary* orthonormal basis B' , $Tr(A) = \sum_{j \in B'} \langle j | A | j \rangle$. (*Hint*: Completeness relation)

¹The X , Y , Z matrices are known as Pauli matrices and H is known as the Hadamard matrix.

²The state $|\Phi^+\rangle$ is known as a Bell state.

Problem 4

A) Given the operators $M_0 = |0\rangle\langle 0|$, $M_1 = |0\rangle\langle 1|$, do they form a valid set of measurement operators? Why/Why not? If so, does the set constitute a projective measurement? Why/Why not?

B) What about $M_0 = |0\rangle\langle +|$, $M_1 = |1\rangle\langle -|$?

C) What about $M_0 = |0\rangle\langle 0|$, $M_1 = |+\rangle\langle +|$?

Problem 5

For a mixed state given by the density matrix ρ , show that $\text{Tr}(\rho^2) \leq 1$, with equality when ρ is a pure state. (*Hint:* You can solve this either by considering ρ as a statistical mixture of some ensemble of states $\{p_i, |\psi_i\rangle\}$ so $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$, or simply using the properties of the density matrix, i.e. that it is hermitian, positive semi-definite and that $\text{Tr}(\rho) = 1$)